

Problem Set 8 Answers

1. There are 120 equally likely outcomes. Thus:

$$P(X = 1) = 25/120 = 5/24$$

$$P(X = 2) = 23/120$$

$$P(X = 3) = 21/120 = 7/40$$

$$P(X = 4) = 19/120$$

$$P(X = 5) = 17/120$$

$$P(X = 6) = 15/120 = 1/8$$

2a. We have $P(X \leq 3) = \sum_{x=1}^3 (1/2)^x = 1/2 + (1/2)^2 + (1/2)^3 = 7/8$.

2b. We have $P(X \leq 10) = \sum_{x=1}^{10} (1/2)^x = 1/2 + (1/2)^2 + (1/2)^3 + \cdots + (1/2)^{10} = 1 - (1/2)^{10} = 1023/1024$.

2c. We have $P(X \leq x) = \sum_{i=1}^x (1/2)^i = 1/2 + (1/2)^2 + (1/2)^3 + \cdots + (1/2)^x = 1 - (1/2)^x = (2^x - 1)/2^x$.

3a. The probability is $\sum_{y=1}^{\infty} (1/2)^{2y} = (1/2)^2 + (1/2)^4 + (1/2)^6 + \cdots = (1/2)^2(1 + (1/2)^2 + (1/2)^4 + \cdots) = (1/2)^2/(1 - (1/2)^2) = 1/3$.

3b. The probability is $\sum_{y=1}^{\infty} (1/2)^{3y} = (1/2)^3 + (1/2)^6 + (1/2)^9 + \cdots = (1/2)^3(1 + (1/2)^3 + (1/2)^6 + \cdots) = (1/2)^3/(1 - (1/2)^3) = 1/7$.

3c. The probability is $\sum_{y=1}^{\infty} (1/2)^{4y} = (1/2)^4 + (1/2)^8 + (1/2)^{12} + \cdots = (1/2)^4(1 + (1/2)^4 + (1/2)^8 + \cdots) = (1/2)^4/(1 - (1/2)^4) = 1/15$.

3d. The probability is $\sum_{y=1}^{\infty} (1/2)^{ry} = (1/2)^r + (1/2)^{2r} + (1/2)^{3r} + \cdots = (1/2)^r(1 + (1/2)^r + (1/2)^{2r} + \cdots) = (1/2)^r/(1 - (1/2)^r) = 1/(2^r - 1)$.

4. We compute $P(X > r) = \sum_{x=r+1}^{\infty} (3/5)^{x-1}(2/5) = (2/5)(3/5)^r/(1 - 3/5) = (3/5)^r$.

5. We compute

$$\begin{aligned} P(a \leq Y \leq b) &= \sum_{y=a}^b (2/7)^{y-1}(5/7) \\ &= (5/7)((2/7)^{a-1} + (2/7)^a + (2/7)^{a+1} + \cdots + (2/7)^{b-1}) \\ &= (5/7)((2/7)^{a-1} + (2/7)^a + (2/7)^{a+1} + \cdots + (2/7)^{b-1})(1 - 2/7)/(1 - 2/7) \\ &= (2/7)^{a-1} - (2/7)^b \end{aligned}$$